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ORIGINAL RESEARCH

CAV1—A Susceptibility Gene for Atrial Fibrillation: The Impact of Coding and Noncoding Variants

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BACKGROUND: Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia with a strong genetic predisposition. Genome-wide association studies have highlighted CAV1 (caveolin 1), a caveolar protein involved in various signaling pathways, as a candidate for cardiac conduction disorders.

METHODS: We explored the role of CAV1 in AF in various models to dissect possible disease mechanisms. First, *CAV1* expression was examined together with the AF risk gene *SHOX2* in a porcine model of induced AF. Then we screened a cohort of 282 patients with early-onset AF to identify genetic variants within *CAV1* and found 1 coding and 5 noncoding variants. The coding variant was functionally investigated in zebrafish, and a comprehensive analysis panel was applied to investigate the noncoding variants.

RESULTS: In the porcine AF model, *CAV1* and *SHOX2* were significantly downregulated in the right atrium and atrioventricular node. Cardiac-specific overexpression of the coding variant in zebrafish increased heart rate and caused fibrillatory waves and loss of the PR interval, supporting a pathogenic effect. Four of the 5 novel identified noncoding variants showed an association with AF and PR interval in published data sets, including 1 with genome-wide significance. The noncoding variants localized to binding sites of transcription factors EOMES, RFX5, TEAD4 and MAX. Luciferase reporter gene assays demonstrated that 3 variants significantly altered the ability of those transcription factors to activate reporter gene expression.

CONCLUSIONS: This work underscores *CAV1* as an AF susceptibility gene and highlights the critical role of coding and noncoding variants in AF disease mechanisms.

Key Words: atrial fibrillation ■ caveolin 1 ■ noncoding variants ■ SHOX2

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia with an estimated prevalence of 60 million cases globally. It is caused by trigger ectopy, acting on a potentially proarrhythmogenic substrate thereby leading to uncoordinated contraction of atria and ventricles.[1,2] The PR interval has been shown to be an electrocardiographic predictor of AF, with both prolonged and shortened PR intervals associated with the disease.[3] Although AF is considered to be a complex condition with multiple environmental risk factors involved, there is increasing evidence for a strong genetic component in the development of the disease.

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RESEARCH PERSPECTIVE

What Is New?

- In several animal models, CAV1 (caveolin 1) is implicated in cardiac conduction disorders: it is downregulated in a porcine AF model, and a coding *CAV1* variant leads to a fibrillatory phenotype in zebrafish.
- Several noncoding *CAV1* variants identified in a cohort with early-onset AF show a possible contribution to pathological processes via a regulatory impact.

What Questions Should Be Addressed Next?

- How do the (regulatory) variants alter normal CAV1 function in cardiac conduction?

Nonstandard Abbreviations and Acronyms

AVN	atrioventricular node
NMD	nonsense-mediated decay
SAN	sinoatrial node

Genome-wide association studies (GWAS) have put the spotlight on various genes encoding ion channels, gap junction proteins, signaling molecules, and transcription factors to play a role in its pathogenesis.[4,5] Nevertheless, the genetic basis and complete genetic network underlying AF remain incompletely understood. Previous GWAS data have shown a significant association between AF and several SNPs located in introns or the 3′UTR of the caveolin 1 (*CAV1*) gene.[6–8] *CAV1* has also been associated with PR interval and P wave duration[7,9] and belongs to the top genetic candidate loci associated with AF.[4]

Caveolins are caveolae plasma membrane proteins involved in various signaling pathways. They have the potential to both negatively and positively regulate the traffic in and out of caveolae membranes.[10] Cav1 is 1 of the 3 known Cav proteins and is ubiquitously expressed in all cell types.[11] It is also an inhibitor of the TGF-ß (transforming growth factor beta) signaling pathway involved in cardiac remodeling and has been studied in isolated cardiac fibroblasts from wild type (WT) and *Cav1*$^{-/-}$ mice[12] and in human atrial fibroblasts.[13]

Genetic variation that affects gene function can alter signaling pathways and their regulatory networks. Its diverse involvement in different cellular processes and the previous GWAS data have proposed *CAV1* as a candidate gene for AF. *CAV1* was also identified as *SHOX2* target gene and showed altered expression levels in animal models with atrial arrhythmias due to *Shox2* deficiency, a gene essential for sinoatrial node (SAN) development and cardiac conduction.[14]

Here, we provide genetic and functional evidence that *CAV1* is involved in AF pathogenesis. We show a common deregulation pattern of *SHOX2* and *CAV1* in a porcine model of AF, highlighting their significance in the disease process. Furthermore, we screened an early-onset AF patient cohort to identify variants within the *CAV1* gene, spanning both exonic and intronic regions. One exonic variant was further investigated in a zebrafish model and multiple intronic variants were subjected to an analysis panel to investigate their impact on gene expression and thus to unveil their contribution to the pathogenesis of AF.

METHODS

Data Availability Statement

The patient cohort was recruited for a previous study, detailed patient characteristics can be found there.[15] Data on identified variants has been uploaded to the gene varaint database LOVD (see below).

Patient and Control Cohorts

The patient cohort comprised 282 German individuals with AF initially recruited for a previous study (detailed patient characteristics available there).[15] Only patients with early-onset AF were included, that is, onset of AF before the age of 60 years. The investigated clinical characteristics included demographic parameters (age and sex [74.1% male]), characteristics of AF (63% of patients with paroxysmal AF), family history (19.9% of patients with a family history of AF, 23.4% with a family history of cardiovascular diseases), echocardiography (ejection fraction, left atrial diameter), ECG parameters (PR interval, RR interval, QRS duration, QT interval), and concomitant diseases (coronary heart disease in 12% of patients, history of myocardial infarction in 5% of patients, valvular heart disease in 6% of patients, dilated cardiomyopathy in 8.8% of patients, hyperthyroidism in 3.8% of patients) or additional cardiovascular risk factors (hypertension in 49.5% of patients, diabetes in 10.3% of patients, hypercholesterolemia in 23.4% of patients).[15]

The study was approved by the ethical commission of the Medical Faculty, University of Heidelberg, Heidelberg, Germany (S-104/2010 “Molekulare Grundlagen SHOX2-bedingter Herzfehlbildungen” 17.03.2010) and was performed in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki and its later amendments. Every participant gave written informed consent including the consent to

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use their DNA for genetic analyses before the inclusion in the study.

Data of control cohorts for the comparison of allele frequencies were taken from DZHKomics,[16] a data set that contains approximately 1200 genomes from unrelated individuals from 6 German population cohorts established as control resource. The coding variant (# 6) was compared with controls in gnomAD (European non-Finnish),[17] as it was absent in the DZHKomics data set, indicating it as a rare variant.

Mutational Screening

Polymerase chain reaction (PCR) primers used to amplify each exon of *CAV1* (NM_001753.5) and the flanking intron-exon boundaries are listed in Table S1. PCRs were performed with HotStarTaq DNA Polymerase (Qiagen) using standard conditions. The PCR products were analyzed by gel electrophoresis. Adequate amplification products were selected for purification and Sanger sequencing done by GENEWIZ (Leipzig, Germany). Data on identified variants have been uploaded to the gene variant database LOVD:

https://databases.lovd.nl/shared/variants/0000987865#00004405
https://databases.lovd.nl/shared/variants/0000987866#00004405
https://databases.lovd.nl/shared/variants/0000987872#00004405
https://databases.lovd.nl/shared/variants/0000987873#00004405
https://databases.lovd.nl/shared/variants/0000987874#00004405
https://databases.lovd.nl/shared/variants/0000987877#00004405

Pig Methods

Animal Ethics Statement

Animal experiments were conducted in compliance with guidelines set forth by the US.National Institutes of Health (NIH publication No. 86-23), the European Union Directive 2010/63/EU, and German laws for animal protection. Approval was obtained from the local Animal Welfare Committee (Regierungspräsidium Karlsruhe, reference number G-296/14). The porcine tissue samples analyzed in this study were derived from animals previously published.[18]

Animal Handling and AF-Induction

AF was induced in German Landrace pigs of both sexes (body weight 25–45 kg) by rapid atrial burst stimulation using an implanted cardiac pacemaker (St Jude Medical, St. Paul, MN, USA) a method that has repeatedly been used to induce persistent AF in porcine models.[19–21] After 20 seconds of high-rate atrial pacing at 40 Hz, pacing was paused to evaluate intrinsic rhythm. During AF episodes, burst stimulation was inhibited by an integrated biofeedback algorithm. Burst pacing resumed whenever the pigs returned to sinus rhythm for more than 15 seconds.

Sedation was achieved with azaperone (5 mg/kg, intramuscular [IM]); Elanco, Bad Homburg, Germany], midazolam (1 mg/kg, IM; Hameln Pharma Plus GmbH, Hameln, Germany), and ketamine (10 mg/kg, IM; Zoetis Deutschland GmbH, Berlin, Germany). Anesthesia was maintained with a propofol bolus (1.5 mg/kg, intravenous [IV]), followed by continuous infusion (4–8 mg/kg per h, IV; Fresenius Kabi, Bad Homburg, Germany). For analgesia, buprenorphine (0.02 mg/kg, IV; Bayer Vital GmbH Tiergesundheit, Leverkusen, Germany) was administered. Mechanical ventilation was performed using the Draeger Primus system (Draeger, Lübeck, Germany). Before surgical preparation of the jugular vein, cefuroxime (750 mg, IV; Ratiopharm GmbH, Ulm, Germany) was administered, and anesthesia was maintained with propofol (4–8 mg/kg per h, IV) and isoflurane (0.5–2 vol.%; Baxter Deutschland GmbH, Heidelberg, Germany). Following cannulation of the right jugular vein, radiofrequency ablation of the distal atrioventricular node (AVN) near the bundle of His was performed under electrocardiographic and fluoroscopic guidance to prevent tachycardia-induced heart failure during AF. Dual-chamber pacemaker systems (St Jude Medical, St. Paul, MN, USA) were implanted as reported earlier[18] and ventricular backup stimulation was provided via the implanted pacemakers. During the 14-day follow-up period, clinical examinations and 4-lead surface ECGs were performed daily. Pigs were housed under specific pathogen-free conditions at 20 °C with a maximum housing density according to directive 2010/63/EU, and a 12/12 hours light/dark cycle. Water was available ad libitum, and pigs were fed twice daily with balanced complete feed (SAF 130M, ZG Raiffeisen, Karlsruhe, Germany). Environmental enrichment was provided with biting woods, chains, and feeding balls. At the end of the 14-day experimental period, hearts were explanted following euthanasia by IV injection of 40 mL potassium chloride (1 mol/L) under deep anesthesia. Tissue samples from 4 distinct regions of the right and left atrium were individually dissected from 7 pigs with AF and 5 controls. RNA expression data and statistical analyses are presented as pooled results, combining the data from AF pigs versus control, from each of the specific 4 tissue regions. Additional samples for sinoatrial node (SAN), the compact AVN immediately proximal to the site where the AVN ablation close to the His bundle was performed, and right and left atrial appendage were dissected individually and processed as a whole from 7 pigs with AF and 5 controls.

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nCounter Expression Analysis

RNA samples from different porcine heart tissues from animals in sinus rhythm (n=5) or with AF (n=7) were analyzed. AF was induced in domestic pigs as described. mRNA expression levels were investigated by nCounter expression analysis at the nCounter Core Facility Heidelberg using the nCounter SPRINT Profiler. This RNA quantification technology allows target measurement from only 50 ng of input with high sensitivity and specificity. Detailed probe design is given in Table S2. The workflow is described at http://www.nanostring.com/elements/workflow. The nCounter customized codeset included 6 negative controls for background substraction, 6 positive controls for lane to lane and cartridge to cartridge normalization, 6 reference genes (*ACTB*, *IPO8*, *RPL4*, *CASQ1*, *HPRT1*, *SDHA*) for mRNA normalization and 2 genes of interest (*CAV1* and *SHOX2*). Normalization of data was performed using the nSolver Analysis Software 4.0 (NanoString Technologies). Most stable expressed reference genes (*IPO8*, *RPL4*, *HPRT1*) were chosen for normalization based on the NormFinder algorithm.[22]

Western Blot

For protein isolation, frozen tissue samples were homogenized using a Tissue Ruptor (QIAGEN, Hilden, Germany) in a lysis buffer containing 150 mM NaCl, 1 mM EDTA, 50 mM Tris–HCl, 0.5% NP-40, 0.25% sodium deoxycholate, 1 mM Na_3VO_4, 1 mM NaF (pH 7.4), and a protease inhibitor cocktail (Complete mini; Roche, Indianapolis, IN, USA). Protein concentrations were determined using a bicinchoninic acid protein assay (Thermo Fisher Scientific) according to the manufacturer's instructions. Equal amounts of protein were diluted in Laemmli sample buffer (Bio-Rad, Munich, Germany) containing 5% beta-mercaptoethanol and boiled for 5 minutes.

Immunodetection of Cav-1 protein was performed after SDS gel electrophoresis, followed by wet transfer to nitrocellulose membranes as previously described.[23] In brief, membranes were blocked in a solution containing 3% bovine serum albumin and 5% dry milk, then incubated with primary antibodies targeting Cav-1 (1:900; ab18199, Abcam, Cambridge, UK) and appropriate horseradish peroxidase-conjugated secondary antibodies (Donkey antirabbit horseradish peroxidase, NA934V, 1:3000, GE Healthcare, Munich, Germany).

Protein bands were detected using enhanced chemiluminescence reagents (ECL Immunoblotting Reagents, GE Healthcare, Buckinghamshire, UK) and quantified using ImageJ 1.45 Software (National Institutes of Health, Bethesda, MD, USA). For normalization, protein content was adjusted to GAPDH levels, using primary antibodies (mouse monoclonal anti-GAPDH, G8140-11, 1:10000, US Biological, Swampscott, MA, USA) and corresponding secondary antibodies (m-IgGκ BP-horseradish peroxidase, sc-516 102, 1:3000, Santa-Cruz Biotechnology, Heidelberg, Germany). The optical density of protein bands was analyzed to determine relative protein expression.

Zebrafish Methods

Generation of a Zebrafish Model

Human WT CAV1 complementary DNA was subcloned together with a myl7/cmlc2-promotor into the pDestTol2CG2 vector that expresses green fluorescent protein under the control of the myl7/cmlc2-promotor to generate an expression plasmid for injection in zebrafish embryos (TE4/6 WT strain), as described previously.[15,24] The *CAV1* mutation (h*CAV1* c.54C>G) was introduced with the QuickChange II Site-Directed Mutagenesis Kit according to the manufacturer's instructions. All constructs were confirmed by sequencing. For overexpression experiments, 25 ng/μL of pDestTol2CG2-*CAV1* WT or pDestTol2CG2-*CAV1* mutant was microinjected into 1-cell-stage embryos. Both morphological analysis and heart rate were determined at 72 hours post fertilization. For the morphological analyses, zebrafish embryos were anesthetized in buffered tricaine methane sulfonate MS-222 (stock 4 mg/mL, dilution used 0.15 mg/mL). For euthanasia, zebrafish embryos were treated with 0.2 mg/mL MS-222.[25]

Functional Analyses in Zebrafish

ECG in zebrafish were performed as previously described.[23,26] Briefly, 2-minute ECG was recorded in 72 hours post fertilization old larvae WT control (injected with KCl) and larvae injected with pDestTol2CG2-*CAV1* WT, or pDestTol2CG2-*CAV1* mutant, which were anesthetized with tricaine methane sulfonate MS-222 (stock 4 mg/mL, dilution used 0.15 mg/mL) to inhibit skeletal muscle movements. For recording of cardiac electrical activity, a 2 μm patch clamp pipette (World Precision Instruments), that was filled with 3M potassium acetate solution, was placed on the pericardial surface between atrium and ventricle by a micromanipulator under microscopic control. A chlorinated silver wire was incorporated in the micropipette to conduct electrical signals to a differential amplifier (EXT10-2F Amplifier, NPI Electronics). Another silver wire was placed in close proximity of the zebrafish larva in the surrounding medium and served as a reference electrode. Voltage signal was recorded (2 minutes recording) in DC mode with the high pass filter set at 0.1 Hz and digitalized

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by PowerLab 4/35 device (AD Instruments). Analysis of the ECG signal was performed using LabChart 7.0 software (AD Instruments).

Quantitative PCR

Following the instructions 700 ng total RNA was transcribed into cDNA using the SuperScript III First-Strand Synthesis Supermix Kit from Invitrogen. For quantitative reverse-transcription PCR, 2 µL of cDNA and 18 µL of Master Mix (10 µL 2× qPCRBio SYBR Green Mix Lo ROX, 0.4 µL Primer fw/rev, 7.6 µL H_2O) was used. Quantitative PCR was conducted on the Quant Studio 3 System. All samples were measured in duplicates. Primers are listed in Table S3.

Cloning of CAV1 Promotor and Mutagenesis

3 kb of the WT *CAV1* promotor including the sites of variants were cloned into the pGl4.17 vector (Promega) using the InFusion Snap Assembly Kit (Takara Bio) according to the manufacturer's instructions. Variants were introduced using the QuickChange Lightning Site-directed Mutagenesis Kit (Agilent) or the Q5 site-directed Mutagenesis Kit (NEB). Primer design and reaction setup was performed as described in the manuals. For transformation, ultracompetent XL-Gold cells were used according to the manufacturer's instructions. Plasmids were isolated using the ZymoPURE Plasmid Miniprep Kit; Sanger Sequencing to confirm mutagenesis was performed by Genewiz.

HEK293TN Cell Culture

Cultivation and Passaging of HEK293TN Cells

HEK293TN cells were obtained from System Biosciences and maintained in DMEM supplemented with 10% FBS and 1% Pen/Strep at 37 °C in a humidified atmosphere. Upon confluency, cells were passaged by rinsing with prewarmed PBS twice followed by an incubation with 0.05% Trypsin–EDTA for 5 minutes until cells were detached. Trypsinisation was stopped by adding culture media and cells were centrifuged at 300×g for 5 minutes. Supernatant was removed and cells were resuspended in culture media. A fraction was transferred to a new flask or cells were counted and the desired amount was seeded.

Transfection of HEK293TN Cells for Luciferase Assay

Next, 6×10^5 HEK293TN cells were seeded into poly-L-lysine coated 24-well plates the day before transfection with 100 ng of the pGl4.74 vector (*Renilla* luciferase) and 200 ng of the other vectors in the corresponding combinations (transcription factor + *CAV1* promotor WT/variant with Firefly luciferase) were transfected in triplicates using the PEI transfection reagent. Cells were harvested 24 hours after transfection in passive lysis buffer. Plates were shaken for 15 minutes at room temperature and then frozen at −80 °C for at least 1 hour.

Dual Luciferase Assay

Frozen plates containing the lysate of transfected cells were thawed and cells were scraped off using a cell scraper. Then, 50 µL of cell lysate per transfection were given into 1 well of a white 96-well plate. Measurements were performed using the Berthold Luminometer and the Dual-Luciferase Assay protocol (Promega) with 25 µL as output volume for LAR II and Stop & Glow Reagent. Briefly, 25 µL of LAR II solution were given to the well and firefly luciferase activity was measured immediately. Next, 25 µL Stop & Glow Reagent were added and *Renilla* luciferase activity was measured as an internal control. Firefly luciferase signals were normalized to *Renilla* control luciferase signals for analysis.

Statistical Analysis

Patient Cohorts

Comparison of genotype frequencies and test for deviation from the Hardy–Weinberg equilibrium were performed using the online tool for case control studies from the Helmholtz Zentrum Munich. Frequencies of genotypes were compared in a minor allele dominant model. Single-nucleotide variant frequencies in the patient cohort and controls retrieved from DZHKomics or gnomAD were tested for deviation from Hardy–Weinberg equilibrium to evaluate whether the observed genotype distributions conformed to expected population genetic patterns. Differences in allele or genotype frequencies between patients and controls were assessed using Armitage's trend test, which is specifically designed to detect linear trends in proportions across ordered genotype categories. A P value <0.05 was considered as statistically significant.

Statistical Analysis of Zebrafish Data

Differences in heart rate between hCAV1 WT and hCAV1 C54G injected embryos were calculated with an unpaired t test, differences in PR interval were calculated with a Kruskal–Wallis test.

Statistical Analysis of Luciferase Assays

Differences of reporter activity were analyzed via 2-way ANOVA corrected for multiple comparisons with the Holm–Šídák approach. A P value <0.05 was considered as statistically significant.

Online tools and databases used (Data S1).

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RESULTS

CAV1 Shows a Common Deregulation Pattern With the AF-Associated Gene *SHOX2* in an AF Porcine Model

Investigation of AF in large animal models, such as pigs, has gained significance as an in vivo model, as it closely resembles human heart physiology in terms of dimensions, structures and functions.[27] Here, after 14 days of induced AF in pigs via right atrial burst stimulation, tissue samples of right and left atrium, the SAN, AVN, and right and left atrial appendage were analyzed. Gene expression analysis of these different cardiac regions using the nCounter methodology revealed a significant downregulation of *CAV1* in right atria and atrioventricular node (Figure 1A through 1F) in pigs with induced AF. Similarly, on protein level, we detected a significant CAV1 downregulation in right atrium and AVN (Figures 1G and 1H). This is an expression pattern that we also observed for the AF-associated gene *SHOX2*, which is reduced in right atria and atrioventricular node. Of note, expression and downregulation of *SHOX2* have been previously reported in SAN tissues in other models but not so far in the atrioventricular node. The common pattern of deregulation of *SHOX2* and *CAV1* in right atria and atrioventricular node in the AF porcine model highlights their probable interplay, underscoring that *CAV1* similarly as *SHOX2* might contribute to processes linked to AF.

Screening of a Patient Cohort With Early-Onset AF Reveals 6 Variants Within the *CAV1* Gene

In a patient cohort (n=282) of early-onset AF before the age of 60,[15] the *CAV1* gene was screened for potential disease-associated variants. Sequencing of its 3 coding exons and adjacent intronic regions revealed 8 variants: 5 intronic variants between exon 1 and exon 2 (variants # 1–5), and 1 variant in the coding region of exon 2 (variant # 6). Of patients with AF onset below the age of 30, 40% carried a noncoding *CAV1* variant, in the age groups between 30 and 40, 40 and 50, and 50 and 60 it was 17.2%, 15.5%, and 15.1%, respectively (Figure S1). Table 1 summarizes the allele counts of the different variants. Allele frequencies were compared with controls from DZHKomics,[16] a data set that contains approximately 1200 genomes from unrelated individuals from 6 German population cohorts. The coding variant (# 6) was compared with controls in gnomAD (European non-Finnish),[17] as it was absent in the DZHKomics data set, indicating it as a rare variant. This variant shows an association with myocardial infarction and AF in the Cardiovascular Disease Knowledge Portal[28] (Table 1).

To test for differences in allele frequencies of the variants in the patient cohort versus the controls, Armitage's trend test was used. One of the variants was significantly enriched in the patient cohort (# 1; Table 1). Associations of the variants with cardiac traits were determined for each variant from the Cardiovascular Disease Knowledge Portal.[28] Notably, 4 variants show an association with PR interval, for 1 variant (# 3) it is genome-wide significant (Table 1). Three of the variants are even associated with AF (# 3, # 4, # 6), confirming an involvement of *CAV1* in cardiac rhythmicity. The full list of phenotype associations of the identified variants can be found in Table S4. Using the Combined Annotation Dependent Depletion (CADD) score, the deleteriousness of the single nucleotide variants were further investigated.[29] A score >10 implies that a variant is among the 10% most deleterious substitutions; a score >20 indicates that a variant is likely to be among the 1% most deleterious substitutions in the human genome. The exonic variant shows a CADD score >20, 1 variant has a score of 17 (# 5) and 3 of the noncoding variants have a score around 10 (# 1, 2, 4), suggesting a possible regulatory effect. These predictions and associations made the variants interesting candidates for further investigation, as they might play a role in the disease mechanisms of AF.

The Identified Coding Variant Increases Heart Rate and Leads to AF in a Zebrafish Model

The coding variant c.54C>G leading to an amino acid change from isoleucine to methionine was predicted to be damaging (Table 1). To assess the particular impact of this *CAV1* variant on cardiac function in vivo, we overexpressed mutant and wild type human *CAV1* in embryonic zebrafish hearts (Figure S2). Therefore, 10 ng/µL expression vectors that express green fluorescent protein under the control of the myl7/cmlc2-promotor together with WT and mutant human *CAV1* were microinjected into zebrafish embryos at the 1-cell-stage and heart rate measurements were performed 72 hours post fertilization. Zebrafish embryos injected with WT *CAV1* expression vectors showed a heart rate of 131.4 beats per minute (n=35). By contrast, the heart rate in zebrafish overexpressing c.54C>G mutant *CAV1* was significantly increased to 141.85 beats per minute (n=39; P=0.0018) (Figure 2B). Neither the overexpression of WT nor mutant *CAV1* had a noticeable effect on the development and morphology of the embryonic heart, providing evidence for the functional significance of CAV1 in heart rate regulation and the particular and pathologic impact of the c.54C>G *CAV1* variant on heart function in vivo (Figure 2, Figure S2).
To also investigate the effect of mutant *CAV1* on the electrophysiology of the zebrafish heart, we performed

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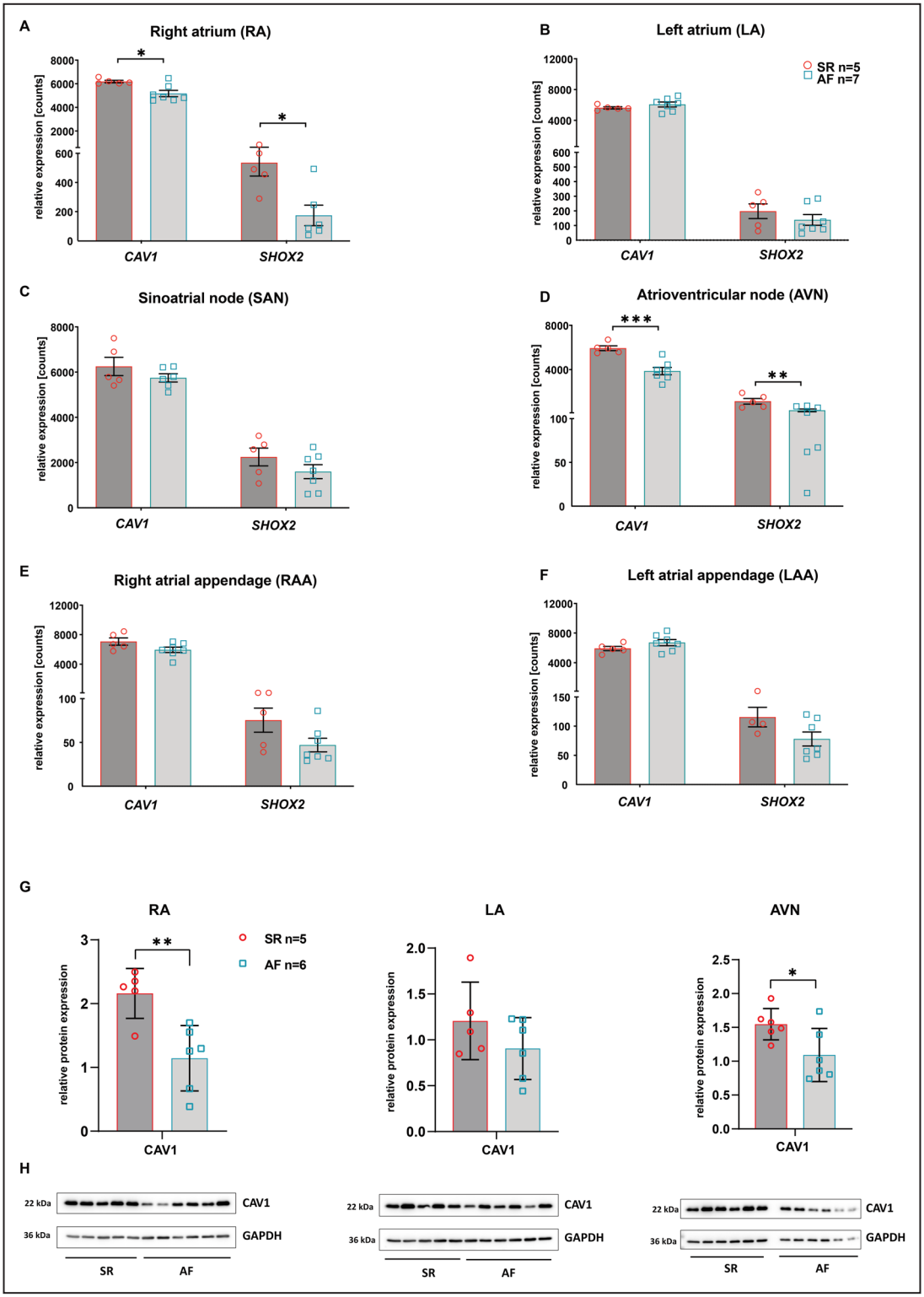

Figure 1. *CAV1* and *SHOX2* expression in a pig model of induced atrial fibrillation compared with sinus rhythm controls.

Gene expression was analyzed using the nCounter technology (**A** through **F**) or Western Blot for protein expression (**G**, **H**). **A**, Right atrium; **B**, Left atrium; **C**, Sinoatrial node; **D**, Atrioventricular node; **E**, Right atrial appendage; **F**, Left atrial appendage. **G**, CAV1 protein expression in RA, LA, and AVN; **H**, Western Blot bands used for expression analysis in **G**. n SR=5; n AF=7/6. *P* values were determined with a *t* test. **P*<0.05; ***P*<0.001; ****P*<0.0001. AF indicates atrial fibrillation; AVN, atrioventricular mode; CAV1, caveolin 1; LA, left atrium; RA, right atrium; and SR, sinus rhythm.

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Table 1. Summary of Localization and Genomic Positions of Identified *CAV1* Variants (GRCh37)

	dbSNP	Localization	Genomic position	Patient number	Allele count	Allele number total	Patient allele frequency	Controls allele frequency	*P* value patient/ctrls	CADD	Association according to Cardiovascular Disease Knowledge Portal
1	rs199578058	7:116 165 159 G>T	Intron	282	2	564	0.0035461	0.00043554	**0.04078**	13.8	
2	rs35697540	7:116 165 160 G>T	Intron	282	10	564	0.0177305	0.00913838	0.0757	10.11	PR interval↓ (*P*=1.40e-3)
3	rs35182456	7:116 165 174 C>T	Intron	282	17	564	0.0301418	0.0274151	0.72442	2.95	**AF (*P*=5.25e-11) (pos.); PR interval↓ (*P*=7.33e-9)**
4	rs45498702	7:116 165 233 C>T	Intron	282	21	564	0.0372340	0.0570557	0.06538	9.61	AF (*P*=7.59e-6) (neg.); PR interval ↑ (*P*=1.85 e-5)
5	rs150496113	7:116 165 237_116 165238insCC	Intron	282	22	564	0.0390071	0.0571055	0.06481	17.1	PR interval ↑ (*P*=1.47e-2)
6	rs150051547	7:116 166 602 C>G (c.54C>G)	Exon	282	1	564	0.0017730	0.00007503 (GnomAD)		21.5	Myocardial infarction (*P*=5.19e-4) (pos.); AF (*P*=1.35e-2) (pos)

Allele count and frequency in the patient cohort (n=282); allele frequencies of controls are from DZHKomics (except for variant 6 with controls from GnomAD European non-Finnish); *P* value was calculated by Armitage's trend test; CADD scores and associated traits as reported by the Cardiovascular Disease Knowledge Portal.[28] *P* values in bold indicate significance, associations in bold indicate genome-wide significance. AF indicates atrial fibrillation; CADD, Combined Annotation Dependent Depletion; dbSNP, Single Nucleotide Polymorphism Database; (neg.) = negative association; (pos.) = positive association; ↓ shortened PR interval; and ↑prolonged PR interval.

ECG recordings (Figure 2C through 2F). Analysis of the ECG, including PR interval measurements of 10 beats per each embryo (n=6/9) revealed the absence of P wave (reported as 0) in 6 out of 9 mutant *CAV1* injected embryos (Figure 2F) with following inability to measure PR interval compared with WT (0.13±0.04 seconds) and *CAV1* WT (0.09±0.04 seconds). In addition, the mutant human *CAV1* injected embryos present fibrillatory (F) waves and absence of sinus rhythm (Figure 2E). Taken together, the absence of PR interval, presence of fibrillatory waves and loss of sinus rhythm are distinctive signs of AF that together with the increased heart rate in the mutant injected zebrafish indicates that the coding c.54C>G *CAV1* mutation severely affects the zebrafish heart.

Characterization of 5 Noncoding Variants Reveals Their Regulatory Impact on Gene Expression

Intronic variants can play a role in disease by influencing gene expression on a regulatory level. The noncoding variants were therefore further investigated. Table 2 shows the predicted effect of the variants on 9 transcripts of *CAV1* by the Ensemble Variant Effect Predictor.[30] Considering the different *CAV1* transcripts, the variants that we identified in the screened population with AF are predominantly intronic or reside in noncoding regions. Only for transcript ENST00000451122.5 (transcript # 9) that has an alternative exon, some of the variants reside within the coding region and lead to a predicted missense mutation (Figure 3A). This transcript was annotated to be targeted for non-sense mediated decay and the GTEx expression profile reports no expression in the cardiac tissues investigated (left ventricle and atrial appendage, Figure 3B). To analyze the presence or absence of the non-sense mediated decay-predicted transcript in cardiac tissue, we used human induced pluripotent stem cell-derived pacemaker cells.[31,32] However, quantitative PCR analysis with transcript-specific primers did not detect expression of this transcript in pacemaker cardiomyocytes (Figure S3). We therefore concluded that transcript ENST00000451122.5 is targeted for non-sense mediated decay and not present in the heart. For all other transcripts (Figure 3B), the identified variants reside in the noncoding or intronic regions on the predicted transcript maps.

The noncoding variants (# 1–5) reside within a region annotated as promotor/enhancer by the GeneHancer database with transcription factor binding ability and may thus alter binding capacities and influence gene expression (Figure 3A). All 5 variants are predicted to reside within the binding sites of several transcription factors including TEAD4, EOMES, MAX and RFX5 among others by the Variant Effect Predictor. TEAD4,

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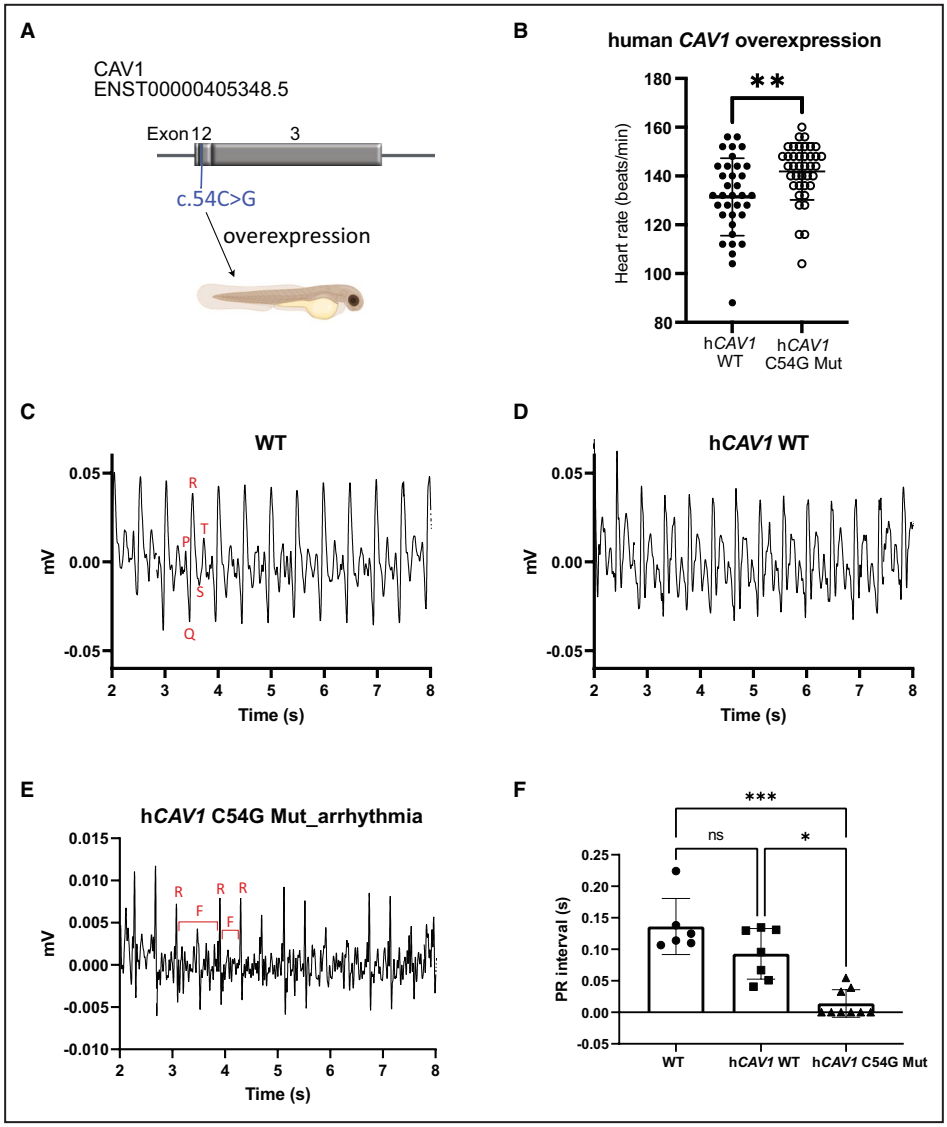

Figure 2. Overexpression of the human *CAV1* c.54C>G variant in zebrafish.
A, ENST00000405348.5 represents a *CAV1* transcript highly expressed in heart tissue. **B**, plasmids carrying *CAV1* WT or *CAV1* mutant were microinjected into one-cell-stage embryos, heart rate was determined at 72 hpf. n (WT)=35, n (CAV1 mutant)=39. *P* value was calculated with an unpaired *t* test;***P*<0.01. **C** and **D**, ECGs on 72 hpf WT and hCAV1 WT embryos are characterized by sinus rhythm and well-defined waves (P, Q, R, S and T). **E**, ECG on 72 hpf mutant CAV1 embryos is characterized by loss of P wave (reported in the graph as 0), presence of fibrillatory (**F**) waves and loss of sinus rhythm. **F**, PR interval measurements show absence of PR interval (reported as 0) in the majority of mutant CAV1 embryos (6 out of 9). n=6/9, mean±SD. *P* value was calculated with a Kruskal–Wallis test, ****P*<0.001;**P*<0.05. CAV1 indicates caveolin 1; hpf, hours post fertilization; and WT, wild type.

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EOMES, MAX and RFX5 have also previously been found to regulate *CAV1* expression in chromatin immunoprecipitation sequencing experiments,[33] making them interesting candidates to investigate if the identified variants in patients may also alter their regulatory influence on *CAV1* gene expression. As we have found similar deregulation patterns between *SHOX2* and its target gene *CAV1* in an induced porcine AF model, we also included *SHOX2* in this list of genes. To investigate if those 5 transcription factors have altered binding capacity associated with the variants and thus lead to a changed activation of transcription, we performed luciferase reporter assays using the promotor sequence containing the respective variant in front of a luciferase reporter gene. Additionally, we included a variant from the same region that has no association with conduction traits and a CADD score of 8.6. This variant showed no change in transactivational activity, further

Table 2. Effect of the Identified Noncoding Variants on the Different *CAV1* Transcripts and on Regulatory Regions as Predicted by the Variant Effect Predictor

	1	2	3	4	5
Transcript	rs199578058	rs35697540	rs35182456	rs45498702	rs150496113
1. ENST00000405348.5	up	up	up	up	up
2. ENST00000393467.1	up	up	up	up	up
3. ENST00000341049.6	intr	intr	intr	intr	intr
4. ENST00000614113.4	intr	intr	intr	intr	intr
5. ENST00000393470.1	intr	intr	intr	intr	intr
6. ENST00000393468.1	up	up	up	up	up
7. ENST00000489856.1	intr, nct	intr, nct	intr, nct	intr, nct	intr, nct
8. ENST00000456473.5	up	up	up	up	up
9. ENST00000451122.5	mis, NMD	mis, NMD	mis, NMD	syn, NMD	fs, NMD
Effect on regulatory region	reg (prom)	reg (prom)	reg (prom)	reg (prom)	reg (prom)
	TF	TF	TF	TF	TF

fs indicates frameshift variant; intr, intron variant; mis, missense variant; nct, noncoding transcript variant; NMD, non-sense mediated decay transcript variant; prom, promotor; reg, regulatory region variant; syn, synonymous variant; TF, transcription factor binding site variant; and Up, upstream gene variant.

supporting the functional impact and specificity of our novel identified variants (Table S5 and Figure S4). Variants # 4 and # 5 always occurred in combination in patients and were investigated together. Luciferase activity was measured after coexpression of the corresponding transcription factors with each variant containing promotor construct. SHOX2, which was included in the analysis based on the common deregulated expression pattern detected in the AF porcine model, indeed was able to activate the reporter, confirming previous data of *CAV1* as a SHOX2 target gene. Except for variants # 4 and # 5, all variants significantly changed reporter gene activation of at least 1 of the transcription factors analyzed (Figure 3C). With variant # 3, RFX5 and SHOX2 binding significantly increased reporter gene expression compared with the WT promotor. The other variants led to a decreased activation of the reporter gene: With variant # 1, EOMES induced a significantly decreased reporter activation. Variant # 2 affected all investigated transcription factors except for TEAD4 and SHOX2, showing highly significant decreased gene activation levels. Together, our results show that these noncoding variants have an impact by altering the binding capacities and activation patterns of transcription factors that are important for gene expression. Thus, they can also play a role in the pathomechanism of *CAV1* deregulation and AF.

DISCUSSION

AF is a disease with a substantial genetic heritability, with many risk genes implicated in its pathogenesis. In this study, we present genetic and functional data of *CAV1* as a gene associated with AF in different models and various approaches. In a porcine AF model, we find a common deregulation pattern of *CAV1* with *SHOX2*, a gene essential for the development of the sinoatrial node and pacemaker function.[34] Significant downregulation of *CAV1* and *SHOX2* was observed in the AVN and in the right atrium, for CAV1 both on mRNA and protein level. Similarly, this has been observed before by Martin et al. who demonstrated that a *CAV1* AF risk allele (rs3807989) is strongly associated with downregulation of *CAV1* in human blood and atrial appendage samples.[35] Together, these data suggest that *CAV1* downregulation is a common finding in AF that is involved in arrhythmicity.

In a patient cohort of early-onset AF, we identified several noncoding and 1 coding variant leading to an amino acid change from isoleucine to methionine. The overexpression of this missense variant in a zebrafish model increased the heart rate. Moreover, although WT and injected human *CAV1* WT ECG traces show a normal heart rhythm including P waves, the *CAV1* mutant presents fibrillatory waves and lacks P waves, a

Figure 3. Impact of noncoding *CAV1* variants on gene expression.
A, Position of *CAV1* variants 1–5 between exon 1 and exon 2 on UCSC Genome Browser (GRCh37). Tracks for *CAV1* transcripts, transcription factor binding, GeneHancer Regulatory Elements (promotor/enhancer) and GeneHancer interactions as well as 100 vertebrate conservation are shown. **B**, Expression of the different *CAV1* transcripts in atrial appendage and left ventricle from Gtex transcript expression browser. **C**, Changed reporter gene activation with variants compared with WT promotor shown in luciferase assays. HEK293TN cells were transfected with TFs and the CAV1 promotor containing the respective variants in front of a luciferase reporter gene. n (WT)=7, n (variants)=3. *P* values were calculated with ANOVA. **P*<0.05; ***P*<0.01; ****P*<0.001. CAV1 indicates caveolin 1; ChIP, chromatin immunoprecipitation sequencing; GH, GeneHancer; NMD, non-sense mediated decay; TF, transcription factor; TPM, transcripts per million; and WT, wild type.

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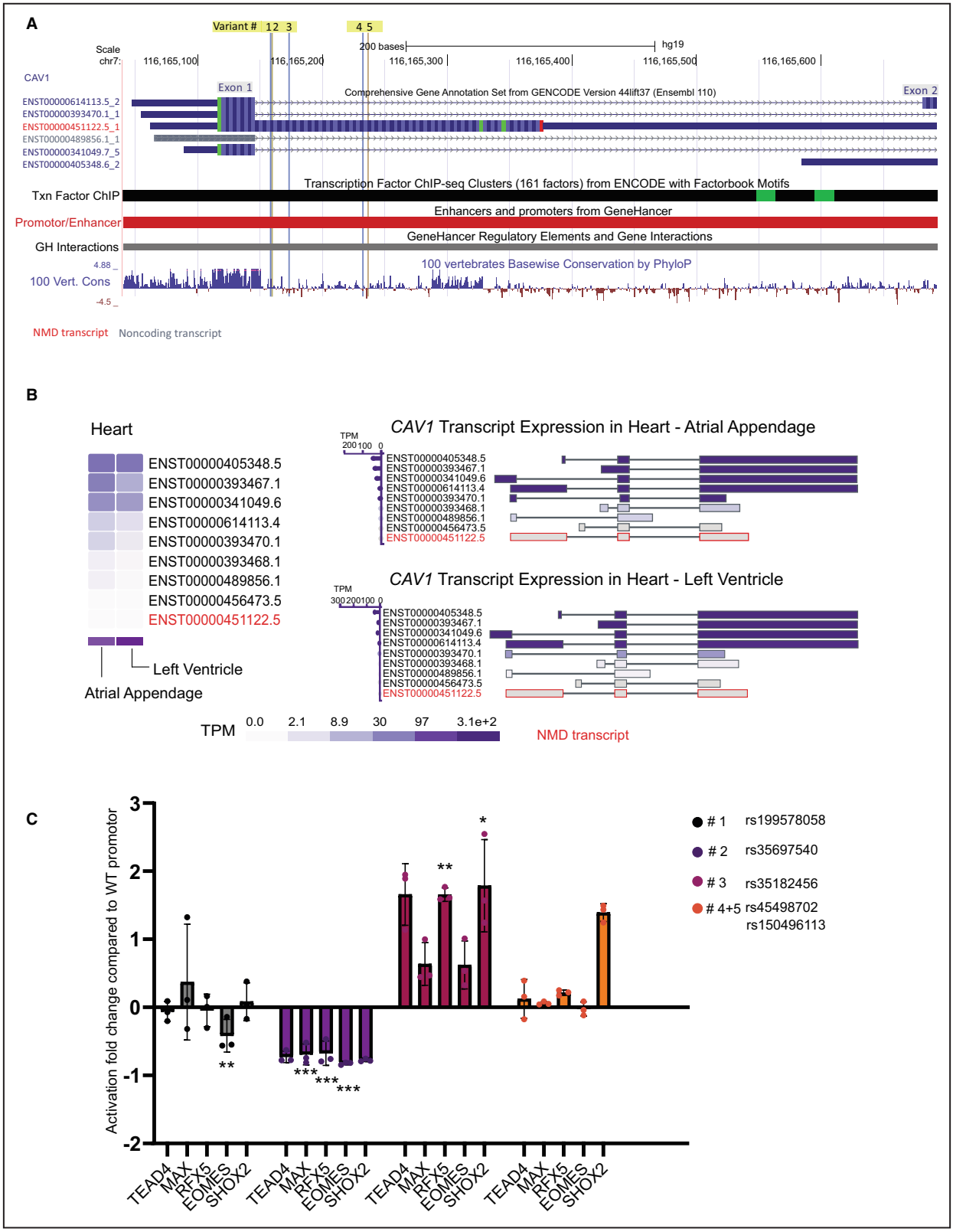

characteristic trait also found in ECGs of patients with AF.[36] These findings directly link the variant to electrophysiological parameters, clearly implying an effect on the disease phenotype. A possible mechanism that could contribute to this effect is proposed by a recent study: Proximity- and affinity-based proteomics have identified aquaporin 1, a water-selective transporting protein, as a CAV1-specific interactor.[37] Aquaporin 1 is

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an important regulator of cardiovascular pathophysiology[38] and involved in pathways like myocardial remodeling and hypertrophy[39] that play a role in arrhythmic circuits. Its interaction with CAV1 in cardiomyocytes could be impaired by the variant, thereby contributing to arrhythmic phenotypes.

Compared with mutations in protein-coding sequences, understanding the function and possible effects of noncoding mutations remains much more difficult. Most GWAS disease and trait-associated variants occur in noncoding regions of the genome. Noncoding variants and their contribution to pathology has been difficult to interpret, in particular due to the lack of understanding when and where these regulatory elements are physiologically active. In the ClinVar database that aims to improve variant interpretation, many noncoding variants are underascertained and more likely to be classified as variants of uncertain significance.[40] Yet, an important role for noncoding variants in disease pathogenesis of AF has been described before: The AFGen Consortium has identified more than 100 genomic loci associated with AF, and most of them reside in noncoding regions of the genome.[41] A noncoding 3′UTR variant associated with AF was also previously identified in the *SHOX2* gene and shown to functionally create a new miRNA binding site, in addition to decreased miRNA plasma levels in these patients [12]. Our data now show that 3 of the identified noncoding *CAV1* variants lead to an altered transcription factor binding and change in gene activation capability. Disruption of transcription factor binding sites due to variants in noncoding regions has been postulated as a disease mechanism.[41] Ye et al., for example, have shown that a functional variant in the region proximal to the *PITX2* gene disrupts enhancer activity and leads to increased AF susceptibility.[42] We predominantly found a reduced activation capability of the transcription factors with the variants; only variant # 3 led to an increased reporter gene activation upon binding of the transcription factors RFX5 and SHOX2. It will be interesting to further find out how increased or decreased gene activation will influence disease mechanisms and signaling networks. The exact relationship between CAV1 and the investigated transcription factors is not known, with the exception of the *Shox2* knockout/knockdown animal models, where deregulated *Cav1* expression has been described.[14] Thus, our data confirm this interaction and proposes a direct mechanism, as one of the variants (# 3) changes the SHOX2 transactivation activity on *CAV1*. This places *CAV1* into a gene regulatory network that is specifically activated by SHOX2 in the SAN, indicating that its deregulation could lead to significant changes in the cardiac conduction system.

Variant # 2 shows the strongest impact on reporter gene activation, as 3 of the 5 investigated transcription factors (EOMES, MAX, and RFX5) are affected. However, it had the lowest CADD score of all variants, which demonstrates the importance to analyze several factors for the prediction of the deleteriousness of noncoding variants. The analysis panel demonstrated here presents a good workflow that includes allele frequencies, association with cardiovascular diseases, CADD score, and further evaluation of regulatory elements to estimate the impact of noncoding variants on disease mechanisms.

Limitations and Strengths of the Study

This study has limitations that should be considered when interpreting the results. The cohort of 282 patients with early-onset AF that was investigated in the current study may be underpowered to detect statistically significant differences in variant frequencies compared with controls. Although 6 candidate variants were identified and further analyzed, not all showed enrichment in the patient cohort or strong associations with arrhythmic phenotypes. A larger sample size would likely enhance statistical power and improve the ability to detect genetic associations. Functional analyses were performed in zebrafish embryos, where the human coding variant was overexpressed specifically in the heart. Variant expression was achieved on top of the endogenous zebrafish *cav1* expression, which can already have an effect, and injection experiments can show high variation due to different injection efficiencies and hence varying dosage. Still, we were able to show a significant impairment of zebrafish embryo heart upon injection, proving that the dosage was high enough without causing any morphological abnormalities (as seen in Figure S2). The transactivational activities of the noncoding variants were assessed in a generic cell model using luciferase reporter assays. These assays lack cardiac tissue specificity, which may limit the physiological relevance of the observed effects on promoter activity. The generation of, for example, induced pluripotent stem cell lines carrying the different variants could provide a more tissue-specific model, as these cells could be differentiated into cardiomyocytes and the impact of the variants could be investigated in a heart specific background. However, it is a time-consuming experiment, in particular if several lines are investigated in parallel.

The strength of this study lies in the application of diverse models and distinct methods to validate the variants. We find associations with CAV1 and arrhythmia in porcine, human and zebrafish backgrounds, showing a conserved role in all these species. By collecting data from various portals and tools, we can show that a single prediction of the deleteriousness of a variant is not sufficient to determine its actual effect, especially but not only for noncoding variants, as also the coding

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variant is not associated with arrhythmic phenotypes yet. Concurrently, this study demonstrates that experimental confirmation of variant effects is indispensable to unveil their full pathogenicity.

CONCLUSIONS

Together, our study confirms previous GWAS findings that *CAV1* gene variants predispose individuals to AF and introduces an analysis panel for non-coding variants that affect regulatory gene function.

ARTICLE INFORMATION

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Disclosure

Data on identified variants has been uploaded to the gene variant database LOVD and can be assessed with the given links. The authors declare no conflict of interest.

Supplemental Material

Tables S1–S5
Figures S1–S4

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